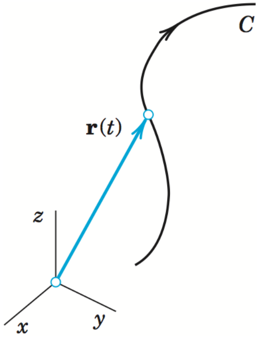
Parametric Curves & Integration Review

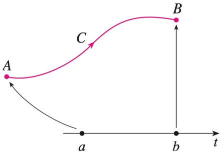
I. Parametric Curves

Defn: A **parametric curve** is a curve in space that depends on a single variable which we called the parameter. In the figure below, t is the parameter.

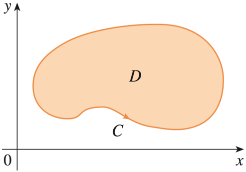


$$\mathbf{r}(t) = [x(t), y(t), z(t)] = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$$

- If t is defined on a finite interval $[a, b]$ then C will be a finite curve.
- If t is defined on a infinite interval, say $[0, \infty]$, then C will be a infinite or closed curve.



finite curve



closed curve

Note: Defining a curve parametrically gives the curve an orientation (direction).

Find **r**(t) by parametrizing the following curves

<u>Ex</u>: $x = y^2 + y - 15,\ z = 0$ (Intersection of)

The easy parameterization is to let

$$\begin{aligned} x &= \\ y &= \\ z &= \end{aligned} \quad \Longrightarrow \quad \mathbf{r}(t) = \qquad t \in$$

<u>Ex</u>: $4x - y = -1,\ 5x - z = 0$ (Intersection of)

Let

$$\begin{aligned} x &= \\ y &= \\ z &= \end{aligned} \quad \Longrightarrow \quad \mathbf{r}(t) = \qquad t \in$$

<u>Ex</u>: $x^2 + y^2 = 2\,, z = -2$ (Intersection of)

Let

$$\begin{aligned} x &= \\ y &= \\ z &= \end{aligned} \quad \Longrightarrow \quad \mathbf{r}(t) = \qquad t \in$$

Ex: $x = y^3 + y,\ z = x - y$ (Intersection of)

The easy parameterization is to let

$$\begin{aligned} x &= \\ y &= \\ z &= \end{aligned} \qquad \Longrightarrow \qquad \mathbf{r}(t) = \qquad\qquad t \in$$

Ex: $x^2 + y^2 = 16$, $z = 2x + y$ (Intersection of)

Let

$$\begin{aligned} x &= \\ y &= \\ z &= \end{aligned} \qquad \Longrightarrow \qquad \mathbf{r}(t) = \qquad\qquad t \in$$

Ex: Find the velocity function given the position function $\mathbf{r}(t) = [1 + t^3, te^{-t}, \sin 2t]$

Recall: *velocity* $= \mathbf{r}'(t) =$

Ex: Let $\mathbf{r}(t) = [\sqrt{t}, 2 - t]$. Find $\mathbf{r}(1)$ and $\mathbf{r}'(1)$

Parameterizing a straight line curves. This is simple if you know:

(1)

(2)

Then the parameterization is given by:

where

<u>Ex</u>: Suppose we know that a line is parallel to $< 1, -1, 0 >$ and passes through $(0, 5, 5)$. Then the parameterization for this line is:

$\mathbf{r}(t) =$ or

<u>Ex</u>: Suppose we instead know that a line passes through the points $(-3, 1, 9)$ and $(-2, 2, 5)$. Then we first find the

and so the parameterization is

$\mathbf{r}(t) =$ or

Parameterizing a straight line segments. In this case we need just the initial and end points of the segment, say P and Q.

Then the parameterization is given by:

$$\mathbf{r}(t) = \qquad\qquad\qquad \text{where}$$

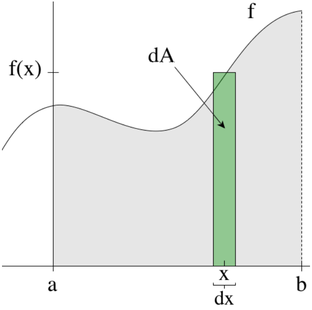
Ex: Suppose we wanted to parameterize a line with initial point $(-1, -1, -9)$ and end point $(10, 0, 15)$. Then the parameterization is given by:

$$\mathbf{v} = \qquad\qquad\qquad\qquad\qquad\qquad \text{where } t \in [0, 1]$$

II. Review: Integration

Remember, integration is another way of saying "add up a lot of small areas to get a whole area". So lets think about it as such:

- Let dA be a small area (of a rectangle) and A the whole area.
- then $dA =$ and so
- $A =$



Note: The limits of integrations (a & b) tell you

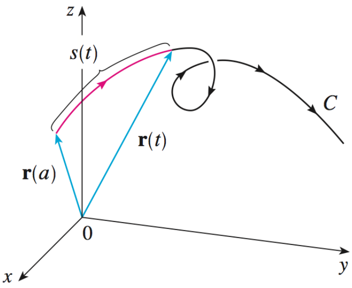
Interpreting Integrals: Identify the physical meaning of the Integral

- $\iiint\limits_E \rho(x,y,z)\,dV$ where E is a solid sphere and $\rho(x,y,z)$ is the density at any point in E

- $\int_a^b f(x)\,dx$ where $f(x)$ is the force required to pull a wagon from a to b

- $\int_C |\mathbf{v}(t)|\,dt$ where $\mathbf{v}(t)$ is the velocity of a particle traveling on a finite curve C

III. Arclength of a Curve



Derive the integral expression for $s(t)$.

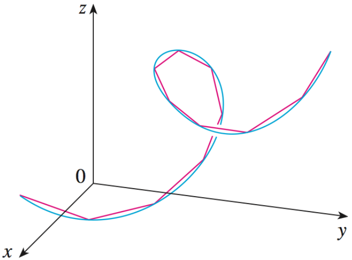
Assume C is parametrized by $\mathbf{r}(t)$ for $a \leq t \leq b$

We want to think of adding up many "small" line segments, so that we can use an integral

$$s = \int_C ds$$

But, what is ds?

Recall: distance = (rate)(time), so

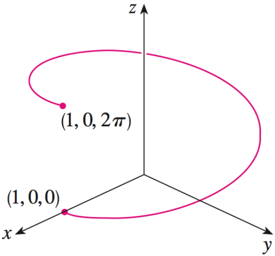


or

Therefore

$s(t) =$

Compute the arclength of the curve: $\mathbf{r}(t) = [\cos t, \sin t, t]$



Need:

Note: